

TEST REPORT

Applicant/Manufacturer: Dragino Technology Co., Limited
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LongCheng Street, LongGang District, Shenzhen China
Report Number: 2501X31094E-EM-02A1

Test Standard (s)

ETSI EN 301 489-1 V2.2.3 (2019-11); ETSI EN 301 489-3 V2.3.2 (2023-01);
ETSI EN 301 489-17 V3.3.1 (2024-09)

Sample Description

Product Type: LoRaWAN Sensor Node
Model No.: DDS75-LB
Multiple Model(s) No.: Refer to Page 2
Trade Mark: DRAGINO
Date Received: 2025/09/26
Issue Date: 2026/01/04

Test Result:	Pass▲
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▲ In the configuration tested, the EUT complied with the standards above.

Prepared and Checked By:

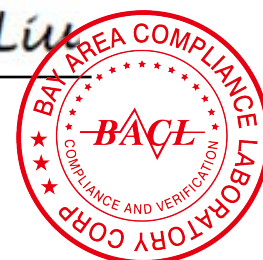
Jason Xiao

Jason Xiao
EMC Engineer

Approved By:

Moon Liu

Moon Liu
EMC Supervisor



Note: The information marked # is provided by the applicant, the laboratory is not responsible for its authenticity and this information can affect the validity of the result in the test report. Customer model name, addresses, names, trademarks etc. are included.
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DOCUMENT REVISION HISTORY

Revision Number	Report Number	Description of Revision	Date of Revision
0	SZ1230802-44707E-EM-02	Original Report	2023/08/29
1	2501X31094E-EM-02A1	Amended Report	2026/01/04

Note:

This is an amended report based on the report SZ1230802-44707E-EM-01, the details as below:

(1) Add model “LDS25-LB, LDS25-LS, LDS25-LB2, CS01-LB, CS01-LS, CS01-LB2, SE01-LS, SE01-LB2, DDS75-LS, DDS75-LB2, DDS45-LS, DDS45-LB2, DDS20-LS, DDS20-LB2, MDS120-LS, MDS120-LB2, SPH01-LS, SPH01-LB2, LMS01-LS, LMS01-LB2, WQS-LB, WQS-LS, WQS-LB2, SE0X-LB, SE0X-LS, SE0X-LB2, WSC2-LB, WSC2-LS, WSC2-LB2, MDS200-LS, MDS200-LB2, LDS40-LB, LDS40-LS, LDS40-LB2, LDS12-LS, LDS12-LB2, DDS04-LS, DDS04-LB2, RS485-LB2, RS485-LS, RS485W-LB, RS485W-LS, RS485W-LB2”.

(2) Upgrading the standard version to “ETSI EN 301 489-17 V3.3.1 (2024-09)”.

Based on above differences, it will affect all test data, so the changed items were performed. The data and photos for original model please refer to the original report.

GENERAL INFORMATION

Product Description for Equipment under Test (EUT)

Product	LoRaWAN Sensor Node
Tested Model	RS485-LS, RS485W-LB2
Multiple Model(s)	LDS25-LB, LDS25-LS, LDS25-LB2, CS01-LB, CS01-LS, CS01-LB2, SE01-LB, SE01-LS, SE01-LB2, DDS75-LS, DDS75-LB2, DDS45-LB, DDS45-LS, DDS45-LB2, DDS20-LB, DDS20-LS, DDS20-LB2, MDS120-LB, MDS120-LS, MDS120-LB2, SPH01-LB, SPH01-LS, SPH01-LB2, LMS01-LB, LMS01-LS, LMS01-LB2, WQS-LB, WQS-LS, WQS-LB2, SE0X-LB, SE0X-LS, SE0X-LB2, WSC2-LB, WSC2-LS, WSC2-LB2, MDS200-LB, MDS200-LS, MDS200-LB2, LDS40-LB, LDS40-LS, LDS40-LB2, LDS12-LB, LDS12-LS, LDS12-LB2, DDS04-LB, DDS04-LS, DDS04-LB2, RS485-LB, RS485-LB2, RS485W-LB, RS485W-LS
Voltage Range	RS485-LS: DC 3.7V from battery RS485W-LB2, RS485-LB2, RS485W-LB: DC 3.6V from battery
Highest operating Frequency [#]	2.48GHz (Provided by the applicant)
Sample serial number	RS485-LS:3AHC-5; RS485W-LB2: 3AHC-7 (Assigned by BACL, Shenzhen)
Sample/EUT Status	Good condition
Adapter Information	N/A
Note: The Multiple models are electrically identical with the test model except for model no., probes, PCB, battery. Please refer to the declaration letter [#] for more detail, which was provided by manufacturer.	

Objective

This test report is in accordance with ETSI EN 301 489-1 V2.2.3 (2019-11), Electromagnetic Compatibility (EMC) standard for radio equipment and services; Part 1: Common technical requirements; Harmonised Standard for Electromagnetic Compatibility. ETSI EN 301 489-3 V2.3.2 (2023-01), Electromagnetic Compatibility (EMC) standard for radio equipment and services; Part 3: Specific conditions for Short Range Devices (SRD) operating on frequencies between 9 kHz and 246 GHz; Harmonised Standard for Electromagnetic Compatibility. ETSI EN 301 489-17 V3.3.1 (2024-09), Electromagnetic Compatibility (EMC) standard for radio equipment and services; Part 17: Specific conditions for Broadband and Wideband Data Transmission Systems; Harmonised Standard for Electromagnetic Compatibility.

The objective is to determine compliance with ETSI EN 301 489-1 V2.2.3 (2019-11), ETSI EN 301 489-3 V2.3.2 (2023-01) and ETSI EN 301 489-17 V3.3.1 (2024-09).

Performance criterion

Performance criterion A

The equipment shall continue to operate as intended without operator intervention. No degradation of performance, loss of function or change of operating state is allowed below a performance level specified by the manufacturer when the equipment is used as intended.

The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.

Performance criterion B

During the application of the disturbance, degradation of performance is allowed. However, no unintended change of actual operating state or stored data is allowed to persist after the test. After the test, the equipment shall continue to operate as intended without operator intervention; no degradation of performance or loss of function is allowed, below a performance level specified by the manufacturer, when the equipment is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level (or the permissible performance loss), or recovery time, is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.

Performance criterion C

Loss of function is allowed, provided the function is self-recoverable, or can be restored by the operation of the controls by the user in accordance with the manufacturer's instructions. A reboot or re-start operation is allowed.

Information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

Measurement Uncertainty

Item	Frequency Range		Expanded Measurement uncertainty
Conducted Emissions	AC Mains	150 kHz ~30MHz	3.66dB(k=2, 95% level of confidence)
Radiated Disturbance	30MHz~200MHz	Horizontal	5.32dB(k=2, 95% level of confidence)
	30MHz~200MHz	Vertical	5.43dB(k=2, 95% level of confidence)
	200MHz~1000MHz	Horizontal	5.77dB(k=2, 95% level of confidence)
	200MHz~1000MHz	Vertical	5.73dB(k=2, 95% level of confidence)
	1GHz~6GHz	/	5.34dB(k=2, 95% level of confidence)

Note: The extended uncertainty given in this report is obtained by combining the standard uncertainty times the coverage factor K with the 95% confidence interval. Otherwise required by the applicant or Product Regulations, Decision Rule in this report did not consider the uncertainty.

Test Facility

The Test site used by Bay Area Compliance Laboratories Corp. (Shenzhen) to collect test data is located on the 5F(B-West) , 6F, 7F, the 3rd Phase of Wan Li Industrial Building D, Shihua Rd, FuTian Free Trade Zone, Shenzhen, China.

Each test item follows test standards and with no deviation.

SYSTEM TEST CONFIGURATION

Description of Test Configuration

The system was configured for testing in a typical fashion (as normally used by a typical user)

For RS485-LS

Test Mode 1: Charging& Lora& Bluetooth transmitting

For RS485W-LB2

Test Mode 2: Lora& Bluetooth transmitting

EUT exercise software

No exercise software was used.

Equipment Modifications

No modifications were made to the EUT.

Support Equipment List and Details

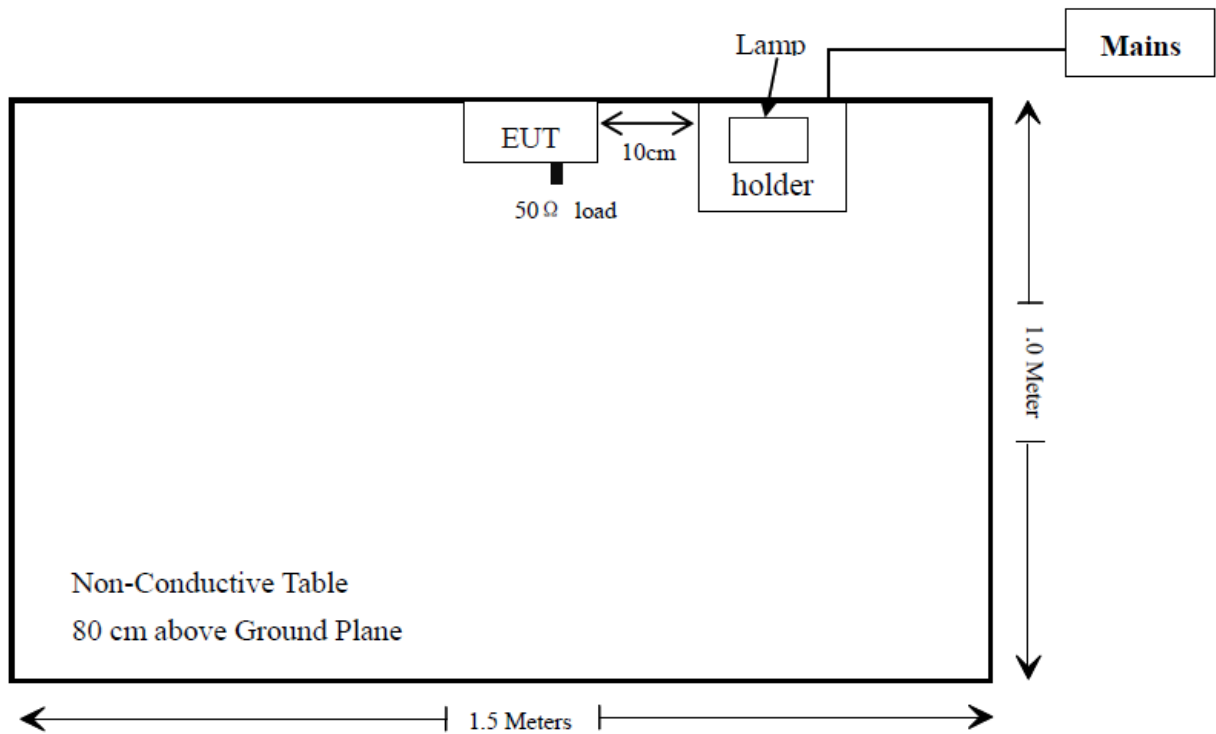
Manufacturer	Description	Model	Serial Number
/	Holder	/	/
/	Lamp	/	/
/	50Ω load	/	/

External I/O Cable

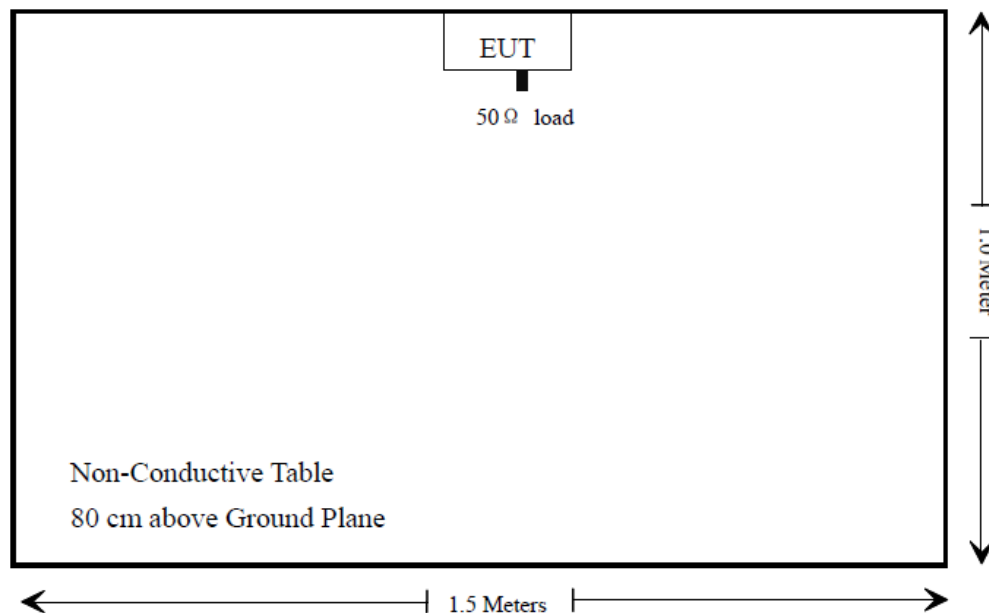
Cable Description	Length (m)	From/Port	To
Un-Shielding Un-Detachable AC Cable	1.2	Holder	Mains

Block Diagram of Test Setup

Test Mode 1



Test Mode 2



SUMMARY OF TEST RESULTS

Rules	Description of Test	Result
§7.1	Reference to clauses EN 301 489-1 §8.4 AC mains power input/output ports	Not Applicable
	Reference to clauses EN 301 489-1 §8.3 DC power input/output ports	Not Applicable*
	Reference to clauses EN 301 489-1 §8.2 Enclosure port of ancillary equipment measured on a stand alone basis	Compliant
	Reference to clauses EN 301 489-1 §8.5 Harmonic current emissions (AC mains input port)	Not Applicable
	Reference to clauses EN 301 489-1 §8.6 Voltage fluctuations and flicker (AC mains input port)	Not Applicable
	Reference to clauses EN 301 489-1 §8.7 Wired network ports	Not Applicable
§7.2	Reference to clauses EN 301 489-1 §9.2 Radio frequency electromagnetic field (80 MHz to 6000 MHz) (EN 61000-4-3)	Compliant
	Reference to clauses EN 301 489-1 §9.3 Electrostatic discharge (EN 61000-4-2)	Compliant
	Reference to clauses EN 301 489-1 §9.4 Fast transients, common mode (EN 61000-4-4)	Not Applicable
	Reference to clauses EN 301 489-1 §9.5 Radio frequency, common mode (EN 61000-4-6)	Not Applicable
	Reference to clauses EN 301 489-1 §9.6 Transients and surges in the vehicular environment (ISO 7637-2)	Not Applicable*
	Reference to clauses EN 301 489-1 §9.8 Surges (EN 61000-4-5)	Not Applicable
	Reference to clauses EN 301 489-1 §9.7 Voltage dips and interruptions (EN 61000-4-11)	Not Applicable

Not Applicable: EUT is battery power supply.

Not Applicable*: Please refer to Applicability overview tables in sections 7.1 and 7.2 of EN 301 489-1 requirements for Radio and ancillary equipment.

TEST EQUIPMENT LIST

Manufacturer	Description	Model	Serial Number	Calibration Date	Calibration Due Date
EMI					
Rohde & Schwarz	EMI Test Receiver	ESR3	102455	2025/09/01	2026/08/31
Sonoma instrument	Pre-amplifier	310 N	186238	2025/09/08	2026/09/07
Sunol Sciences	Broadband Antenna	JB1	A040904-1	2023/07/20	2026/07/19
Unknown	Chamber A Cable	Cable A1	Cable A1	2025/09/08	2026/09/07
Unknown	Chamber A Cable	Cable A2	Cable A2	2025/09/08	2026/09/07
Audix	EMI Test software	E3	19821b(V9)	NCR	NCR
TDK	Chamber	Chamber A	2#	2023/07/12	2026/07/11
Rohde & Schwarz	Spectrum Analyzer	FSV40	101605	2025/09/01	2026/08/31
A.H.System	Preamplifier	PAM-0118P	489	2025/09/08	2026/09/07
Schwarzbeck	Horn Antenna	BBHA9120D(1201)	1143	2023/07/26	2026/07/25
Unknown	Chamber B Cable	Cable B1	Cable B1	2025/09/08	2026/09/07
Unknown	Chamber B Cable	Cable B2	Cable B2	2025/09/08	2026/09/07
Unknown	Chamber B Cable	Cable B3	Cable B3	2025/09/08	2026/09/07
JD	Filter Switch Unit	DT7220FSU	DS79906	2025/08/12	2026/08/11
JD	Multiplex Switch Test Control Set	DT7220SCU	DS79903	2025/08/12	2026/08/11
Audix	EMI Test software	E3	191218 V9(c)	NCR	NCR
TDK	Chamber	Chamber B	1#	2023/07/14	2026/07/13

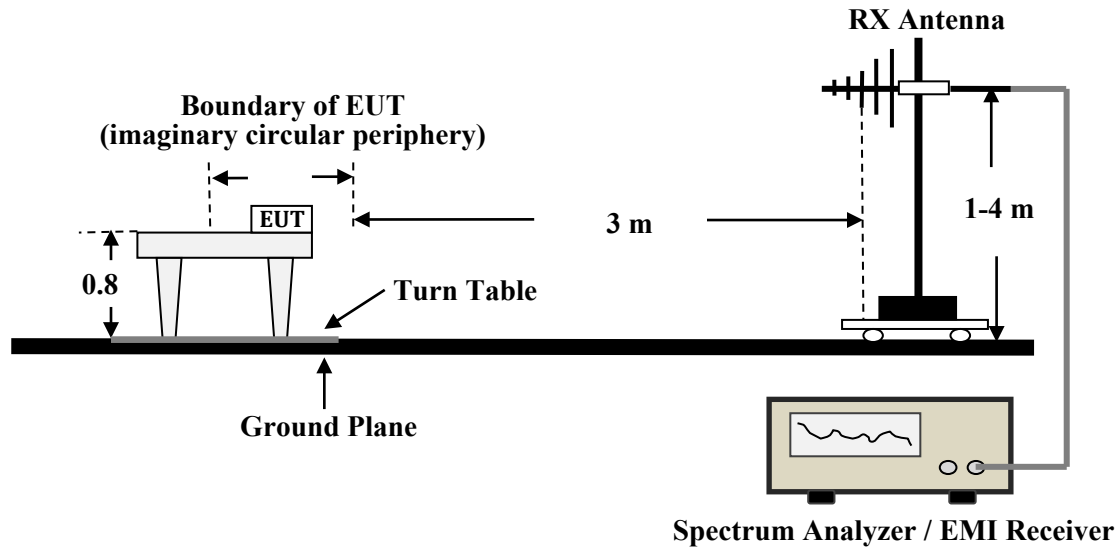
Manufacturer	Description	Model	Serial Number	Calibration Date	Calibration Due Date
EMS					
TESEQ	ESD Generator	NSG 438	1476	2025/07/15	2026/07/14
Agilent	Signal Generator	8665B	3744A01675	2024/12/04	2025/12/03
AR	Power Amplifier	200W1000/M2	H1004497	NCR	NCR
AR	Amplifier	60S1G6	348712	NCR	NCR
AR	Antenna	ATL80M1G	348837	NCR	NCR
AR	Antenna	ATT700M12G	349411	NCR	NCR
BACL	Test Software	VEE PRO	V2.3 VXE	NCR	NCR
EM TEST	AC Source	MV2616	V0939105173	2024/12/04	2025/12/03
THERMO	IMMUNITY TESTER	EMC PRO PLUS	1108237	2025/09/01	2026/08/31

Statement of Traceability: Bay Area Compliance Laboratories Corp. (Shenzhen) attests that all calibrations have been performed in accordance to requirements that traceable to National Primary Standards and International System of Units (SI).

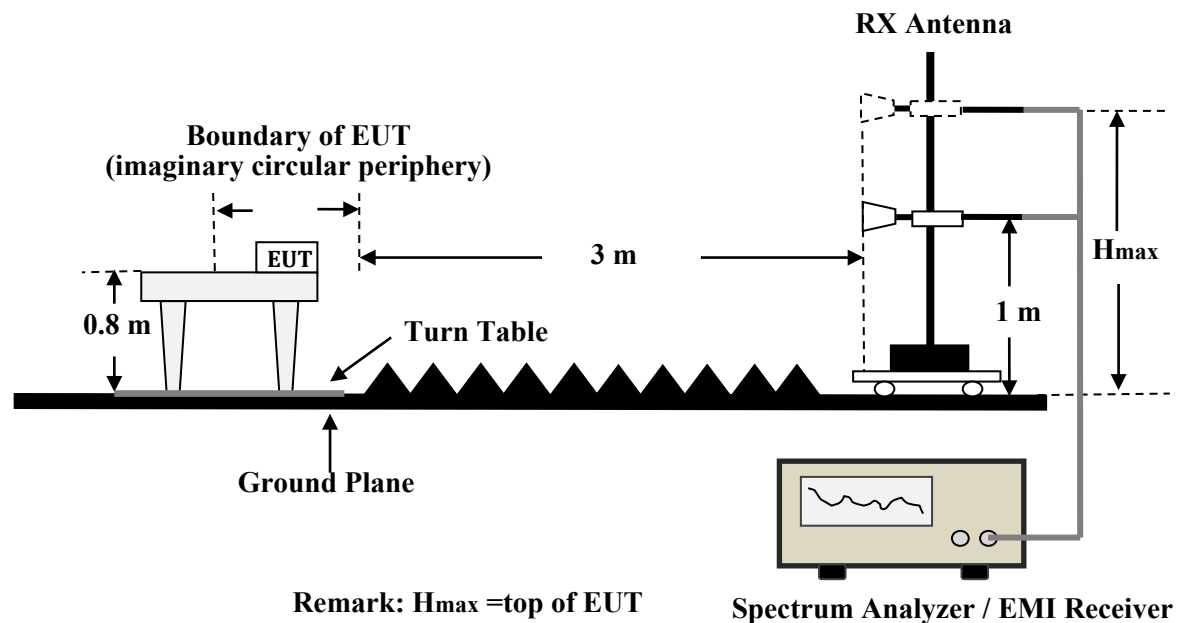
§7.1 - RADIATED EMISSIONS

Test System Setup

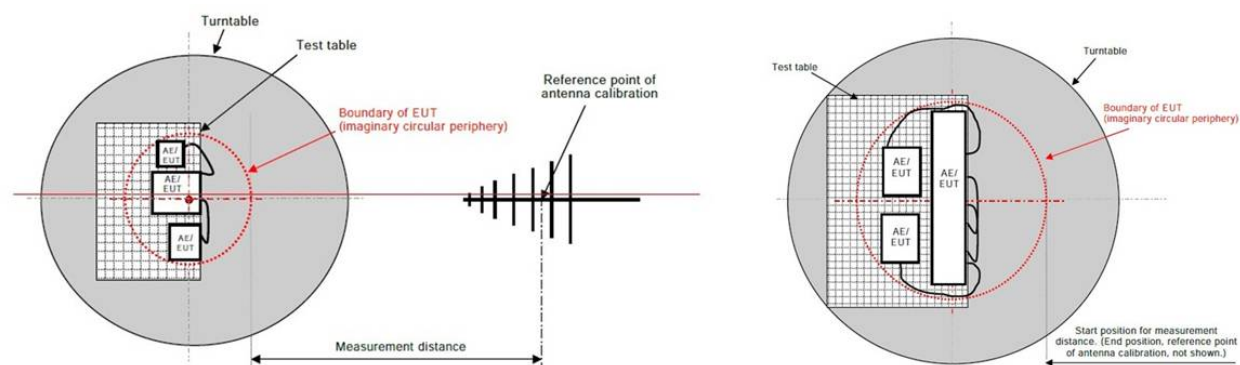
Below 1GHz for Radiated Emissions



Above 1GHz for Radiated Emissions



Radiated Emissions Setup Configuration



The radiated emission tests were performed in the 3 meters chamber test site, using the setup accordance with the CISPR 16-1-4:2012, CISPR 16-2-3:2010. The limit was specified in EN 301 489-1.

The external I/O cables were draped along the test table and formed a bundle 30 to 40 cm long in the middle.

The spacing between the peripherals was 10 cm.

EMI Test Receiver and Spectrum analyzer Setup

During the radiated emission test, the EMI test receiver and spectrum analyzer setup was set with the following configurations:

Frequency Range	RBW	Video B/W	IF B/W	Detector
30 MHz – 1000 MHz	100 kHz	300 kHz	120 kHz	QP
1 GHz – 6 GHz	1 MHz	3 MHz	-	Peak
1 GHz – 6 GHz	1 MHz	10 Hz	-	Peak

Test Procedure

Maximizing procedure was performed on the highest emissions to ensure that the EUT complied with all installation combinations.

If emission level of the EUT in Peak measurement mode is 20dB lower than peak limit line (that means the emission level in Peak measurement mode complies with both Peak and average limit lines) then only Peak measurement result is reported. Otherwise, Emission in average measurement mode shall be measured, and reported for frequency range above 1GHz.

Level & Over Limit Calculation

The Level is calculated by adding the Antenna Factor and Cable Loss, and subtracting the Amplifier Gain from the Read Level. The basic equation is as follows:

$$\text{Factor} = \text{Antenna Factor} + \text{Cable Loss} - \text{Amplifier Gain}$$

$$\text{Level} = \text{Read Level} + \text{Factor}$$

The “**Over limit**” column of the following data tables indicates the degree of compliance with the applicable limit. For example, an Over limit of -6 dB means the emission is 6 dB below the limit for Class B. The equation for margin calculation is as follows:

$$\text{Over limit} = \text{Level} - \text{Limit}$$

Test Data

Environmental Conditions

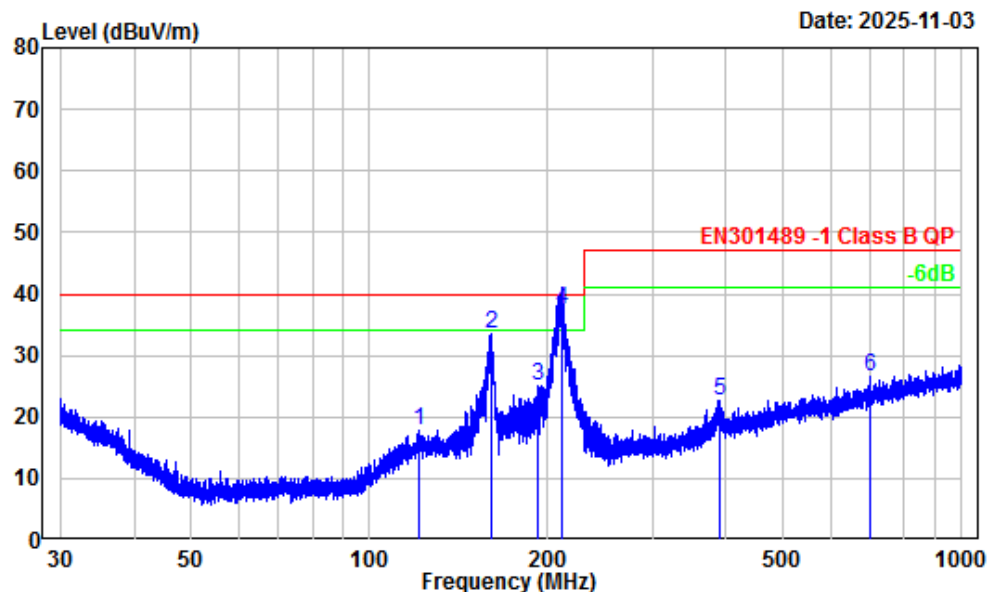
Temperature:	22.3~24.7 °C
Relative Humidity:	51~59 %
ATM Pressure:	101.1~101.6 kPa

The testing was performed by Anson Su on 2025-11-03 for below 1GHz and Leon Guo on 2025-10-30 for above 1GHz.

Test Mode 1

30 MHz~1 GHz

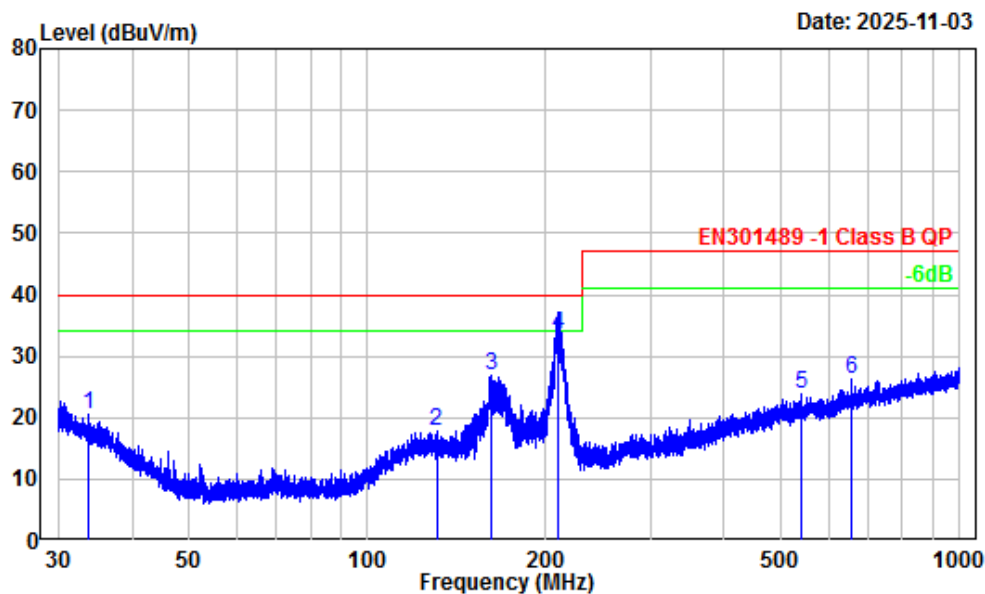
Horizontal



Site : Chamber A
Condition : 3m Horizontal
Project Number : 2501X31094E-EMA1
Test Mode : Mode 1
Detector: Peak RBW/VBW: 100/300kHz
Detector: QP RBW : 120kHz
Tester : Anson Su

	Freq Factor		Read Level		Limit	Over	Remark
	MHz	dB/m	dBuV	dBuV/m	Line	Limit	
1	120.81	-11.37	29.16	17.79	40.00	-22.21	Peak
2	160.63	-12.72	46.11	33.39	40.00	-6.61	Peak
3	192.08	-14.01	39.07	25.06	40.00	-14.94	Peak
4	210.97	-14.06	51.60	37.54	40.00	-2.46	QP
5	391.24	-8.82	31.41	22.59	47.00	-24.41	Peak
6	699.61	-3.52	29.96	26.44	47.00	-20.56	Peak

Vertical

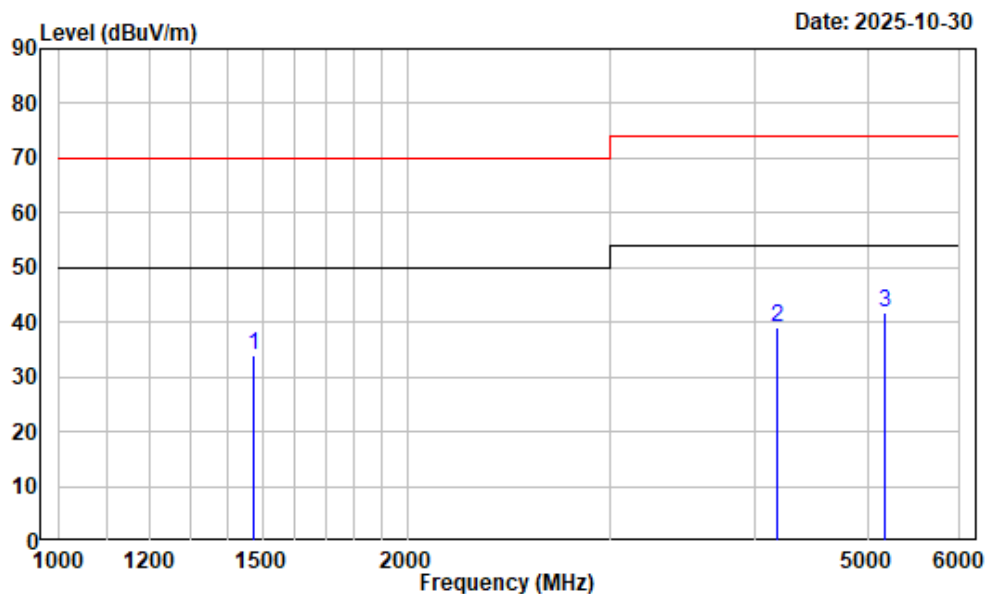


Site : Chamber A
Condition : 3m Vertical
Project Number : 2501X31094E-EMA1
Test Mode : Mode 1
Detector: Peak RBW/VBW: 100/300kHz
Detector: QP RBW : 120kHz
Tester : Anson Su

	Freq Factor		Read Level	Limit Level	Over Limit	Remark
	MHz	dB/m	dBuV	dBuV/m	dBuV/m	dB
1	33.78	-8.13	28.73	20.60	40.00	-19.40 Peak
2	130.72	-11.28	29.20	17.92	40.00	-22.08 Peak
3	162.18	-12.74	39.47	26.73	40.00	-13.27 Peak
4	209.59	-13.98	47.40	33.42	40.00	-6.58 QP
5	541.61	-5.68	29.63	23.95	47.00	-23.05 Peak
6	656.53	-4.00	30.29	26.29	47.00	-20.71 Peak

1-6 GHz

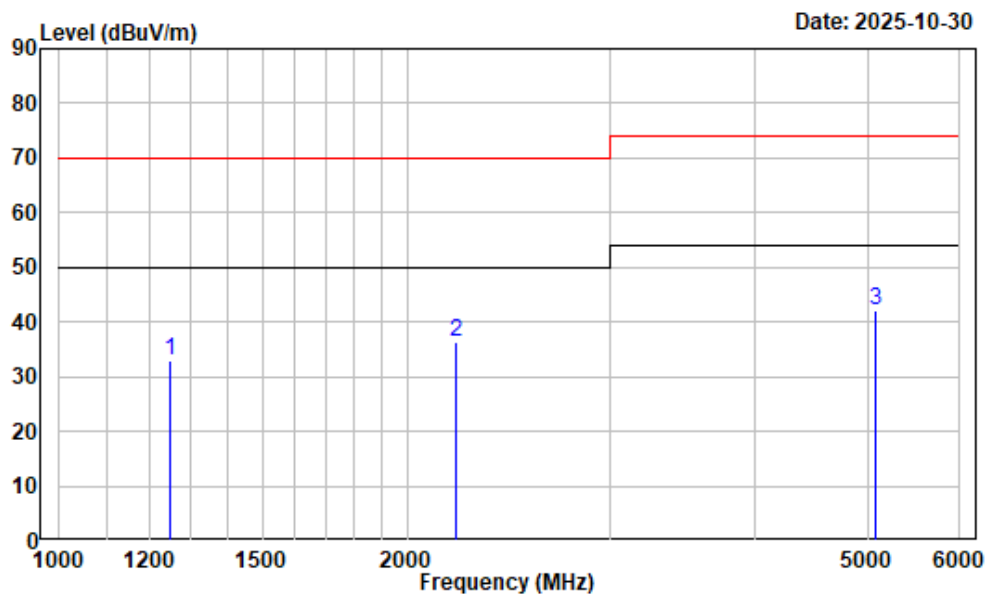
Horizontal



Site : chamber B
Condition : Horizontal
Project Number : 2501X31094E-EMA1
Test mode : Mode 1
Tester : Leon Guo
Spectrum setting: Peak reading: RBW:1MHz VBW:3MHz Detector:Peak

	Freq Factor		Read		Limit	Over	Remark
	MHz	dB/m	Level	Level	Line	Limit	
	MHz	dB/m	dBuV	dBuV/m	dBuV/m	dB	
1	1472.500	-14.21	48.15	33.94	70.00	-36.06	Peak
2	4169.375	-9.41	48.47	39.06	74.00	-34.94	Peak
3	5174.375	-7.26	49.14	41.88	74.00	-32.12	Peak

Vertical



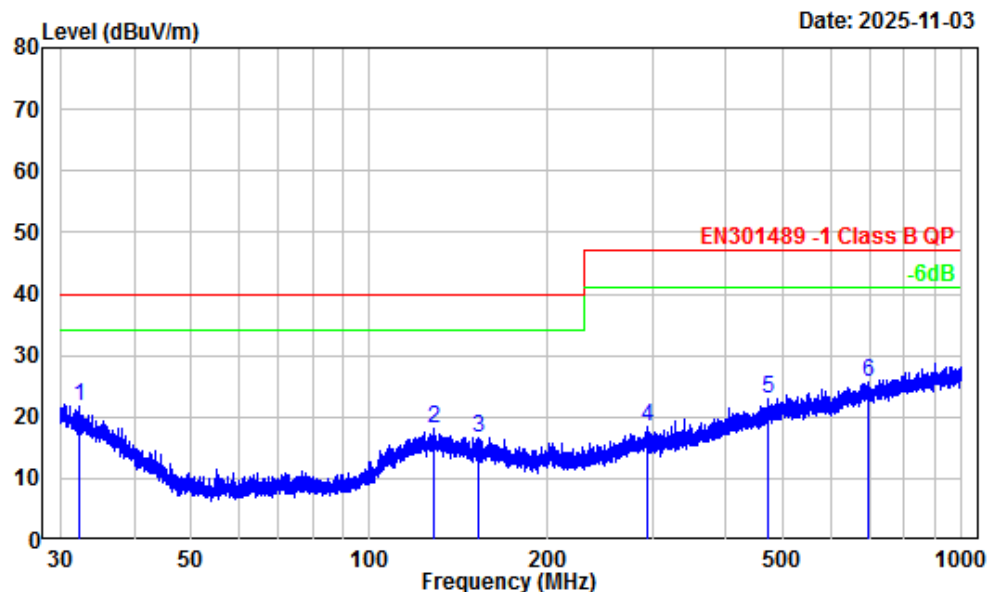
Site : chamber B
Condition : Vertical
Project Number : 2501X31094E-EMA1
Test mode : Mode 1
Tester : Leon Guo
Spectrum setting: Peak reading: RBW:1MHz VBW:3MHz Detector:Peak

	Freq Factor		Read		Limit	Over	Remark
	MHz	dB/m	Level	Level	Line	Limit	
	MHz	dB/m	dBuV	dBuV/m	dBuV/m	dB	
1	1247.500	-14.62	47.61	32.99	70.00	-37.01	Peak
2	2206.875	-10.83	47.01	36.18	70.00	-33.82	Peak
3	5070.625	-7.36	49.63	42.27	74.00	-31.73	Peak

Test Mode 2

30 MHz~1 GHz

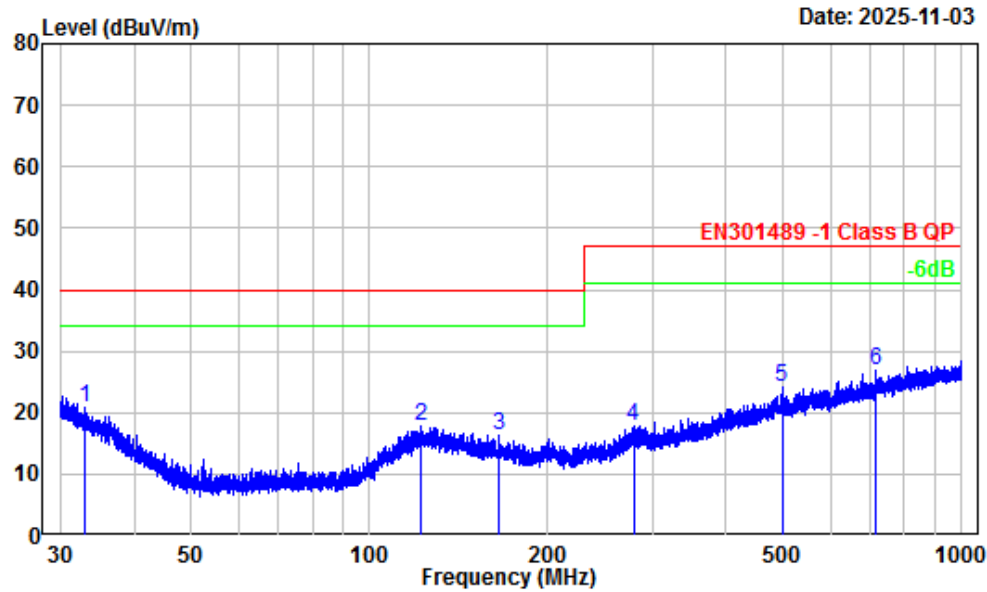
Horizontal



Site : Chamber A
Condition : 3m Horizontal
Project Number : 2501X31094E-EMA1
Test Mode : Mode 2
Detector: Peak RBW/VBW: 100/300kHz
Tester : Anson Su

Freq Factor		Read Level	Level	Limit Line	Over Limit	Remark
MHz	dB/m	dBuV	dBuV/m	dBuV/m	dB	
1	32.38	-7.26	28.93	21.67	40.00	-18.33 Peak
2	128.39	-11.15	29.19	18.04	40.00	-21.96 Peak
3	152.80	-12.55	29.21	16.66	40.00	-23.34 Peak
4	294.11	-11.22	29.54	18.32	47.00	-28.68 Peak
5	470.73	-6.73	29.80	23.07	47.00	-23.93 Peak
6	696.86	-3.53	29.15	25.62	47.00	-21.38 Peak

Vertical

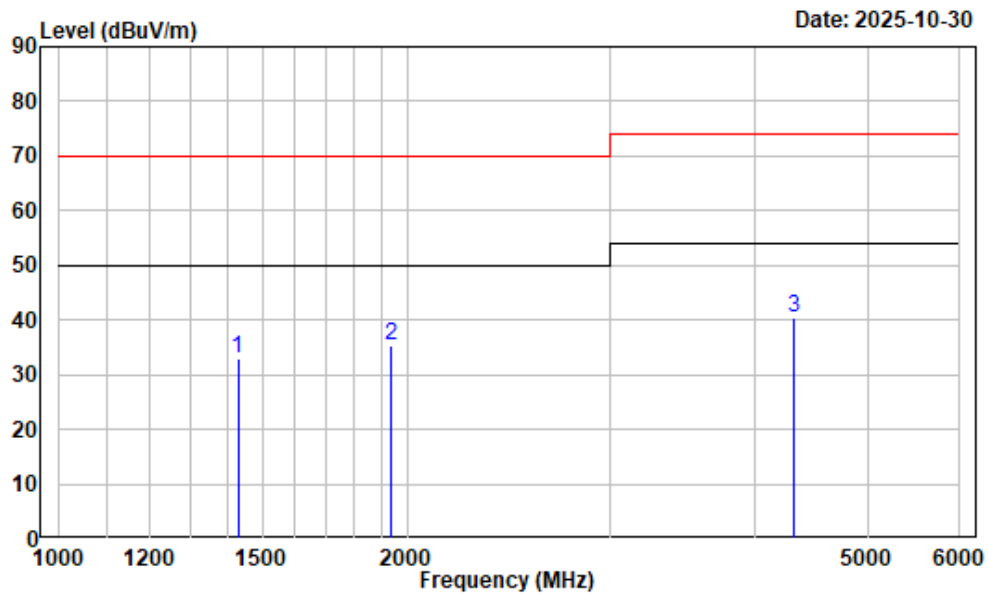


Site : Chamber A
Condition : 3m Vertical
Project Number : 2501X31094E-EMA1
Test Mode : Mode 2
Detector: Peak RBW/VBW: 100/300kHz
Tester : Anson Su

	Freq Factor		Read	Limit	Over	Remark
	MHz	dB/m	Level	Level	Line	
			dBuV	dBuV/m	dBuV/m	dB
1	32.92	-7.54	28.43	20.89	40.00	-19.11 Peak
2	122.19	-11.22	28.91	17.69	40.00	-22.31 Peak
3	165.12	-12.91	29.23	16.32	40.00	-23.68 Peak
4	279.04	-11.25	29.07	17.82	47.00	-29.18 Peak
5	497.24	-5.86	29.97	24.11	47.00	-22.89 Peak
6	715.43	-3.29	30.14	26.85	47.00	-20.15 Peak

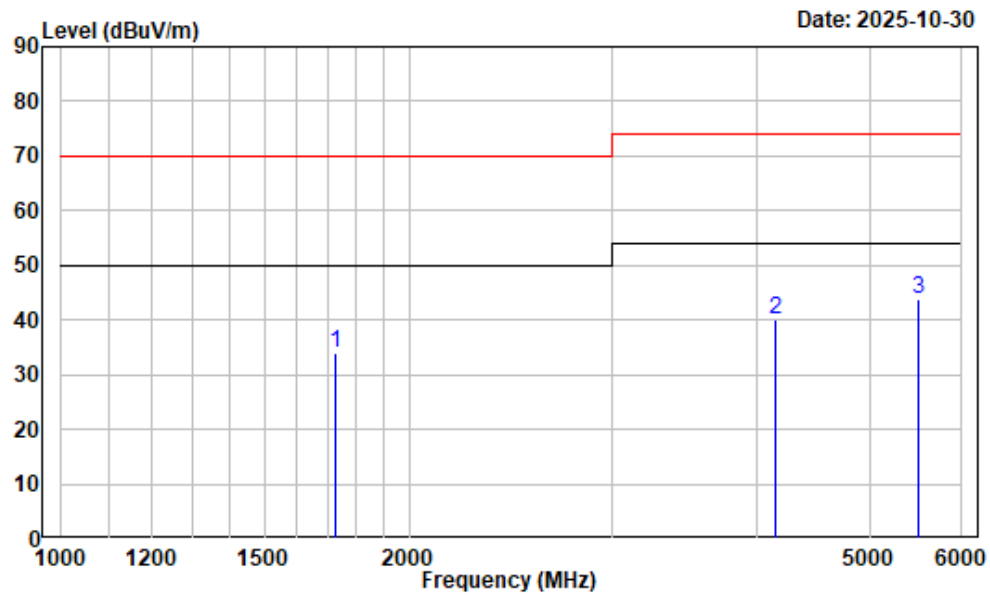
1-6 GHz

Horizontal



Site : chamber B
Condition : Horizontal
Project Number : 2501X31094E-EMA1
Test mode : Mode 2
Tester : Leon Guo
Spectrum setting: Peak reading: RBW:1MHz VBW:3MHz Detector:Peak

	Freq Factor		Read		Limit	Over	Remark
	MHz	dB/m	Level	Level	Line	Limit	
	MHz	dB/m	dBuV	dBuV/m	dBuV/m	dB	
1	1429.375	-14.08	47.17	33.09	70.00	-36.91	Peak
2	1936.250	-13.13	48.39	35.26	70.00	-34.74	Peak
3	4312.500	-8.74	49.21	40.47	74.00	-33.53	Peak

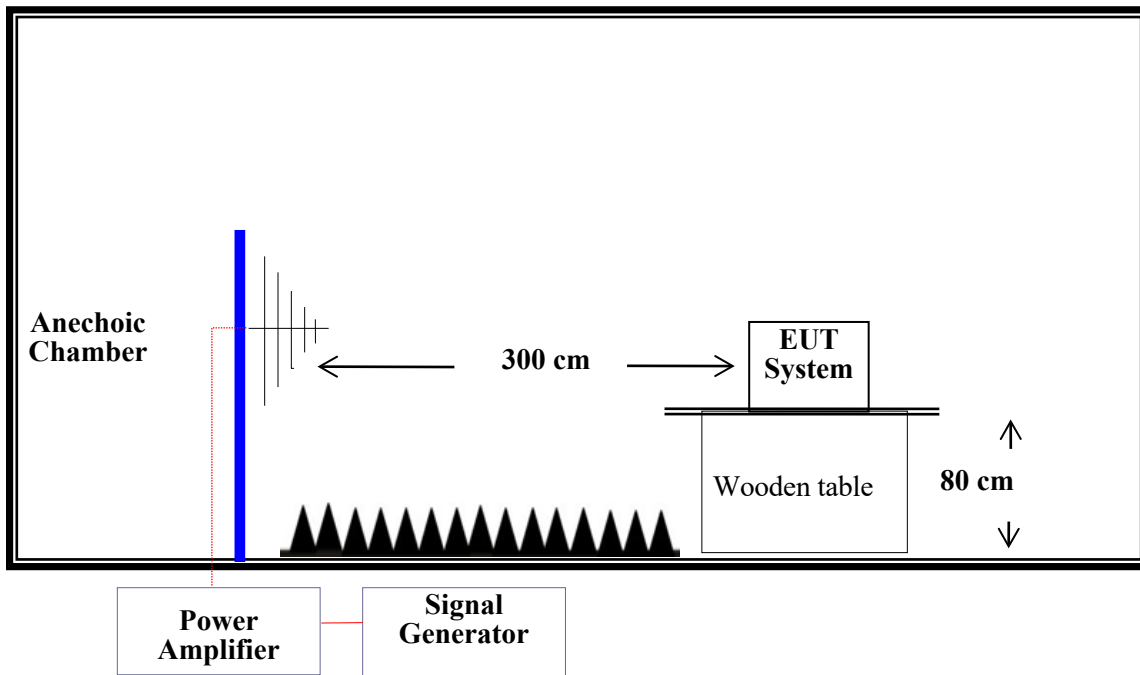
Vertical

Site : chamber B
Condition : Vertical
Project Number : 2501X31094E-EMA1
Test mode : Mode 2
Tester : Leon Guo
Spectrum setting: Peak reading: RBW:1MHz VBW:3MHz Detector:Peak

	Freq Factor		Read		Limit	Over	Remark
	MHz	dB/m	Level	Level	Line	Limit	
	MHz	dB/m	dBuV	dBuV/m	dBuV/m	dB	
1	1728.750	-13.95	47.96	34.01	70.00	-35.99	Peak
2	4139.375	-9.55	49.72	40.17	74.00	-33.83	Peak
3	5515.000	-6.13	49.94	43.81	74.00	-30.19	Peak

§7.2 - RF ELECTROMAGNETIC FIELD (80 MHz to 6000 MHz)

Test System Setup



Test Standard

ETSI EN 301 489-1 V2.2.3 (2019-11) / EN 61000-4-3:2006+A1:2008+A2:2010

Test Level

Level	Field Strength (V/m)
1.	1
2.	3
3.	10
X.	Special

Test Procedure

The EUT and its simulators are placed on a turn table which is 0.8 meter above the ground. The EUT is set 3 meters away from the transmitting antenna which is mounted on an antenna tower. Both horizontal and vertical polarizations of the antenna are set on test. Each of the four sides of EUT must be faced this transmitting antenna and measured individually. In order to judge the EUT performance, a CCD camera is used to monitor the EUT

All the scanning conditions are as follows:

Condition of Test

Remarks

- | Condition of Test | Remarks |
|-----------------------|----------------------|
| 1. Field Strength | 3 V/m (Test Level 2) |
| 2. Radiated Signal | Modulated |
| 3. Scanning Frequency | 80 - 6000 MHz |
| 4. Frequency step | 1% |
| 5. Dwell Time | 1 Sec. |

Test Data and Setup Photo

Environmental Conditions

Temperature:	26.1 °C
Relative Humidity:	43 %
ATM Pressure:	100.9 kPa

The testing was performed by Jason Li on 2025-11-04.

Test Mode 1&2

Modulation: Amplitude 80%, 1 kHz sine wave

Frequency Range (MHz)	Front Side (3 V/m)		Rear Side (3 V/m)		Left Side (3 V/m)		Right Side (3 V/m)	
	VERT	HORI	VERT	HORI	VERT	HORI	VERT	HORI
80-6000	A	A	A	A	A	A	A	A

Note:

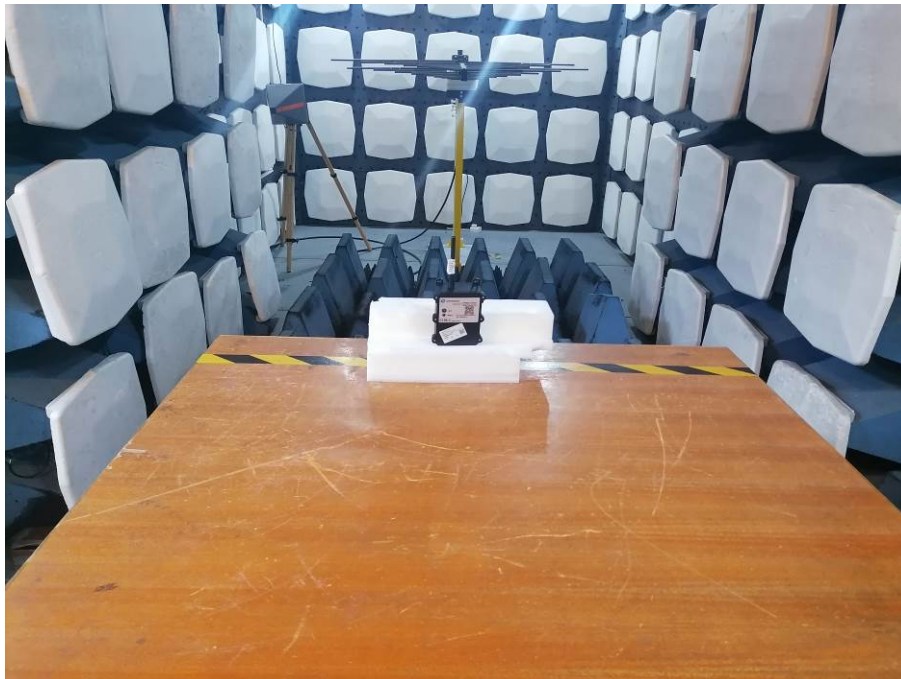
EUT Operation Mode (Lora&BT)

Note: "A" stand for, during test, operate as intended No loss function, no degradation of performance, no unintentional transmissions and after test, no degradation of performance, no loss of function, no loss of stored data or user programmable functions.

Test Mode 1



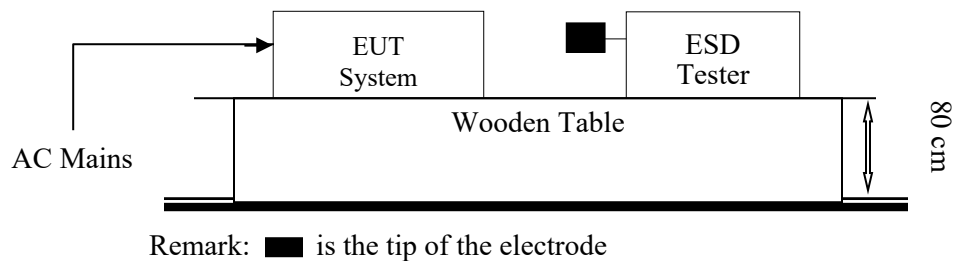
Test Mode 2



Test Setup Photos

§7.2 - ELECTROSTATIC DISCHARGE

Test System Setup



EN 61000-4-2 specifies that a tabletop EUT shall be placed on a non-conducting table which is 80 centimeters above a ground reference plane and that floor mounted equipment shall be placed on a insulating support approximately 10 centimeters above a ground plane. During the tests, the EUT is positioned over a ground reference plane in conformance with this requirement.

For tabletop equipment, a 1.6 by 0.8-meter metal sheet (HCP) is placed on the table and connected to the ground plane via a metal strap with two 470 k Ohms resistors in series. The EUT and attached cables are isolated from this metal sheet by *0.5-millimeter* thick insulating material. A Vertical Coupling Plane (VCP) grounded on the ground plane through the same configuration as in the HCP is used.

Test Standard

ETSI EN 301 489-1 V2.2.3 (2019-11) / EN 61000-4-2:2009

Test Level

Level	Test Voltage Contact Discharge (±kV)	Test Voltage Air Discharge (±kV)
1.	2	2
2.	4	4
3.	6	8
4.	8	15
X.	Special	Special

Performance criterion: B

Test Procedure

Air Discharge:

This test is done on a non-conductive surface. The round discharge tip of the discharge electrode shall be approached as fast as possible to touch the EUT. After each discharge, the discharge electrode shall be removed from the EUT. The generator is then re-triggered for a new single discharge and repeated 10 times for each pre-selected test point. This procedure shall be repeated until all the air discharge completed.

Contact Discharge:

All the procedure shall be same as Section 8.3.1 of EN 61000-4-2, except that the tip of the discharge electrode shall touch the EUT before the discharge switch is operated.

Indirect discharge for horizontal coupling plane

At least 20 single discharges shall be applied to the horizontal coupling plane, at points on each side of the EUT. The discharge electrode positions vertically at a distance of 0.1m from the EUT and with the discharge electrode touching the coupling plane.

Indirect discharge for vertical coupling plane

At least 20 single discharges shall be applied to the center of one vertical edge of the coupling plane. The coupling plane, of dimensions 0.5m × 0.5m, is placed parallel to, and positioned at a distance of 0.1m from the EUT. Discharges shall be applied to the coupling plane, with this plane in sufficient different positions that the four faces of the EUT are completely illuminated.

Test Data and Setup Photo

Environmental Conditions

Temperature:	25.6 °C
Relative Humidity:	40 %
ATM Pressure:	100.9 kPa

The testing was performed by Jason Li on 2025-11-04.

*Test Mode 1***Table 1: Electrostatic Discharge Immunity (Air Discharge)**

IEC 61000-4-2 Test Points	Test Levels							
	-2 kV	+2 kV	-4 kV	+4 kV	-8 kV	+8 kV	-15 kV	+15 kV
Front (2 points)	A	A	A	A	A	A	/	/
Left (1 point)	A	A	A	A	A	A	/	/
Right (1 point)	A	A	A	A	A	A	/	/
Top (2 points)	A	A	A	A	A	A	/	/

Table 2: Electrostatic Discharge Immunity (Direct Contact)

IEC 61000-4-2 Test Points	Test Levels							
	-2 kV	+2 kV	-4 kV	+4 kV	-6 kV	+6 kV	-8 kV	+8 kV
/	/	/	/	/	/	/	/	/

Table 3: Electrostatic Discharge Immunity (Indirect Contact HCP)

IEC 61000-4-2 Test Points	Test Levels							
	-2 kV	+2 kV	-4 kV	+4 kV	-6 kV	+6 kV	-8 kV	+8 kV
Front Side	A	A	A	A	/	/	/	/
Back Side	A	A	A	A	/	/	/	/
Left Side	A	A	A	A	/	/	/	/
Right Side	A	A	A	A	/	/	/	/

Table 4: Electrostatic Discharge Immunity (Indirect Contact VCP)

IEC 61000-4-2 Test Points	Test Levels							
	-2 kV	+2 kV	-4 kV	+4 kV	-6 kV	+6 kV	-8 kV	+8 kV
Front Side	A	A	A	A	/	/	/	/
Back Side	A	A	A	A	/	/	/	/
Left Side	A	A	A	A	/	/	/	/
Right Side	A	A	A	A	/	/	/	/



Note: ➡ represents air discharge, ➡ represents direct contact

Note: The list is only for photos of the location where the discharge can be made, the others not listed are without discharge points, or not the EUT part.

*Test Mode 1***Table 1: Electrostatic Discharge Immunity (Air Discharge)**

IEC 61000-4-2 Test Points	Test Levels							
	-2 kV	+2 kV	-4 kV	+4 kV	-8 kV	+8 kV	-15 kV	+15 kV
Front (2 points)	A	A	A	A	A	A	/	/
Left (1 point)	A	A	A	A	A	A	/	/
Right (1 point)	A	A	A	A	A	A	/	/
Top (2 points)	A	A	A	A	A	A	/	/

Table 2: Electrostatic Discharge Immunity (Direct Contact)

IEC 61000-4-2 Test Points	Test Levels							
	-2 kV	+2 kV	-4 kV	+4 kV	-6 kV	+6 kV	-8 kV	+8 kV
/	/	/	/	/	/	/	/	/

Table 3: Electrostatic Discharge Immunity (Indirect Contact HCP)

IEC 61000-4-2 Test Points	Test Levels							
	-2 kV	+2 kV	-4 kV	+4 kV	-6 kV	+6 kV	-8 kV	+8 kV
Front Side	A	A	A	A	/	/	/	/
Back Side	A	A	A	A	/	/	/	/
Left Side	A	A	A	A	/	/	/	/
Right Side	A	A	A	A	/	/	/	/

Table 4: Electrostatic Discharge Immunity (Indirect Contact VCP)

IEC 61000-4-2 Test Points	Test Levels							
	-2 kV	+2 kV	-4 kV	+4 kV	-6 kV	+6 kV	-8 kV	+8 kV
Front Side	A	A	A	A	/	/	/	/
Back Side	A	A	A	A	/	/	/	/
Left Side	A	A	A	A	/	/	/	/
Right Side	A	A	A	A	/	/	/	/



Note:  represents air discharge,  represents direct contact

Note: The list is only for photos of the location where the discharge can be made, the others not listed are without discharge points, or not the EUT part.

Test Mode 1



Test Mode 2



Test Setup Photos

EXHIBIT A - EUT PHOTOGRAPHS

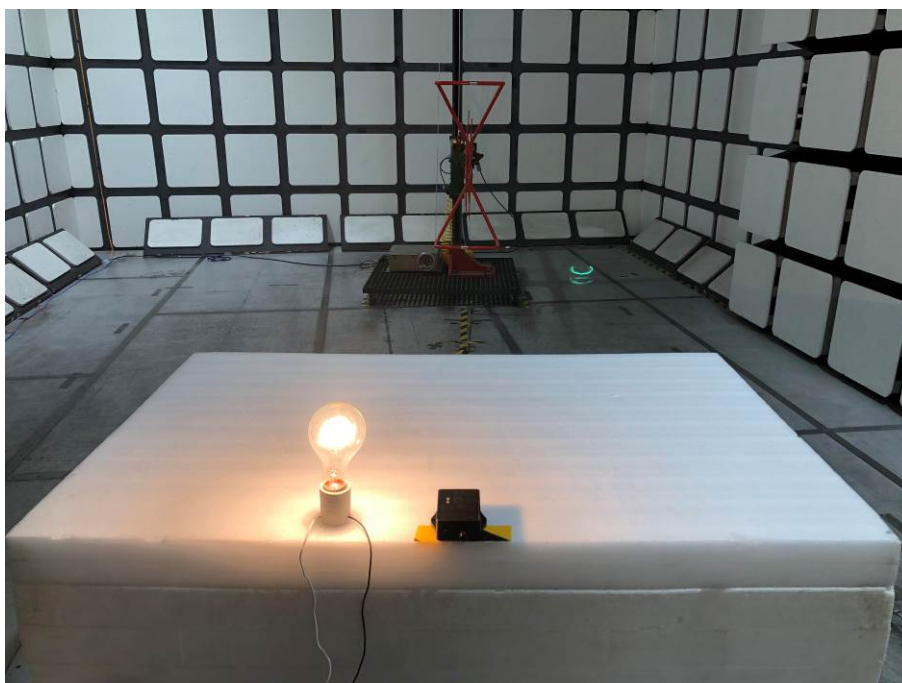
Please refer to the report number is 2501X31094E-EUT.

EXHIBIT B - TEST SETUP PHOTOGRAPHS

Test Mode 1
Radiated Emissions - Front View (Below 1 GHz)



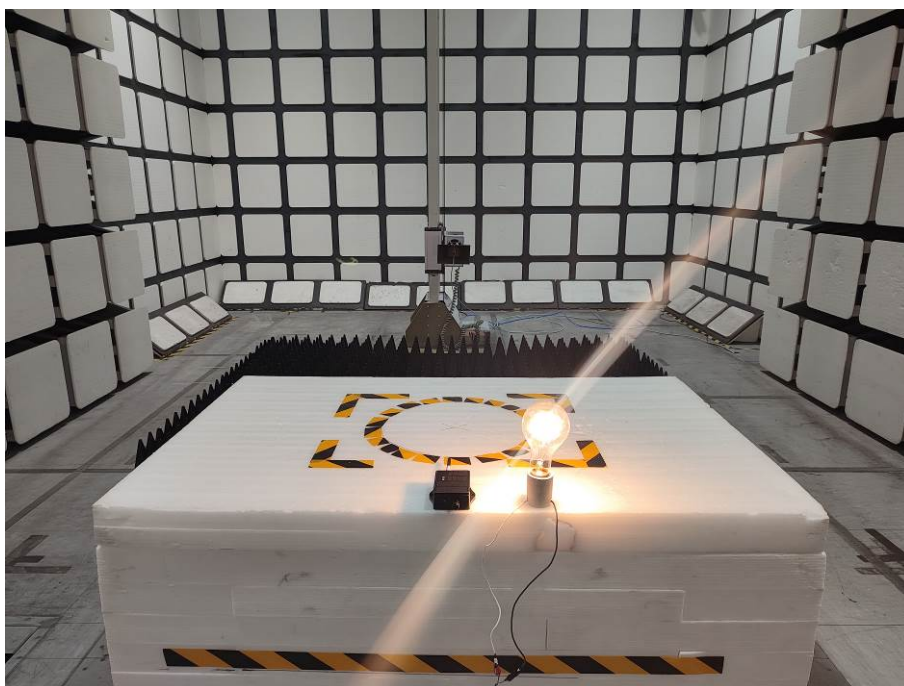
Radiated Emissions - Rear View (Below 1 GHz)



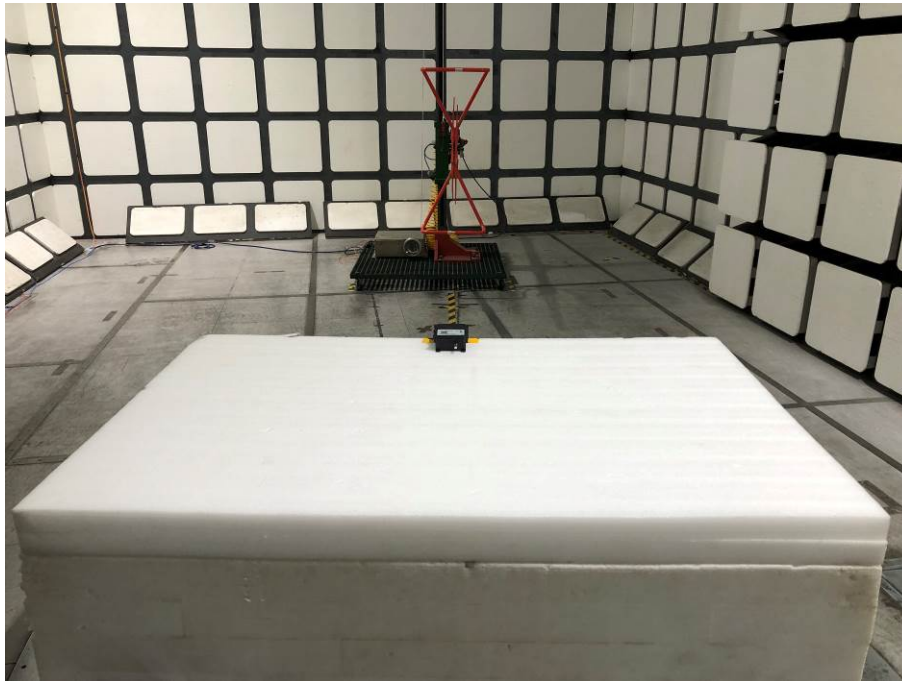
Radiated Emissions - Front View (Above 1 GHz)



Radiated Emissions - Rear View (Above 1 GHz)



Test Mode 2
Radiated Emissions - Front View (Below 1 GHz)



Radiated Emissions - Rear View (Below 1 GHz)



Radiated Emissions - Front View (Above 1 GHz)



Radiated Emissions - Rear View (Above 1 GHz)

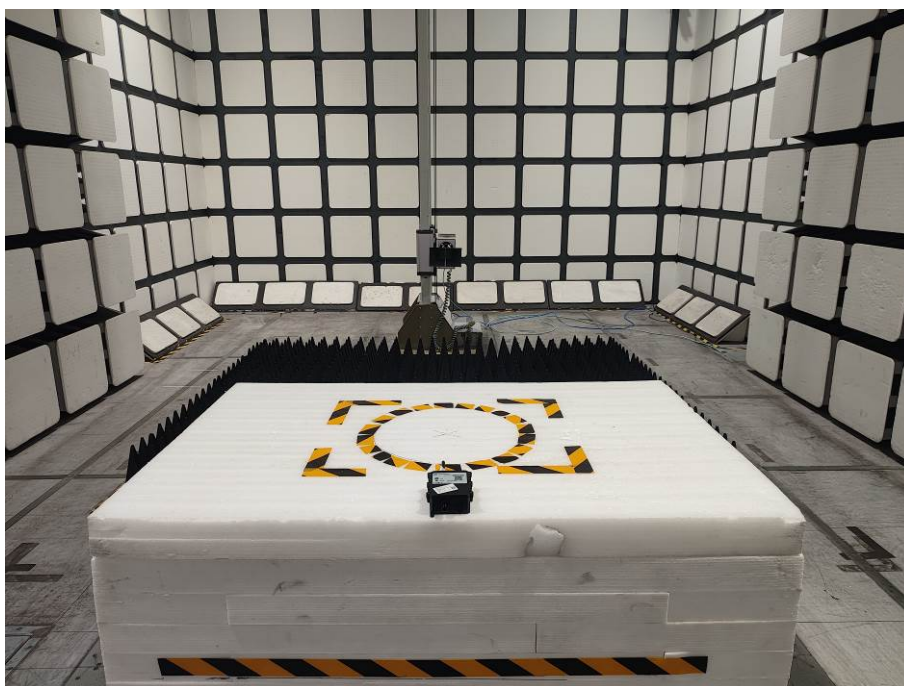


EXHIBIT C - PRODUCT SIMILARITY DECLARATION OF LETTER

Dragino Technology Co., Limited.

Address: Room 202,BaoChengTai industrial park,No.8 CaiYun LongCheng Street,LongGang District,
Shenzhen China

Tel: 86-0755-86610821; Fax: 86-0755-61673660

Email: edwin@dragino.com

Product Similarity Declaration

Date: 2025-11-09

Dear Sir/Madam,

We, **Dragino Technology Co., Limited.** hereby declare that we have a product named as LoRaWAN Sensor Node (model: DDS75-LB),were tested by Bay Area Compliance Laboratories Corp. (Shenzhen), Meanwhile, For our marketing purpose, we would like to list a series model(LDS25-LB, LDS25-LS, LDS25-LB2, CS01-LB, CS01-LS, CS01-LB2, SE01-LB, SE01-LS, SE01-LB2, DDS75-LB, DDS75-LS, DDS75-LB2, DDS45-LB, DDS45-LS, DDS45-LB2, DDS20-LB, DDS20-LS, DDS20-LB2, MDS120-LB, MDS120-LS, MDS120-LB2, SPH01-LB, SPH01-LS, SPH01-LB2, LMS01-LB, LMS01-LS, LMS01-LB2, WQS-LB, WQS-LS, WQS-LB2, SE0X-LB, SE0X-LS, SE0X-LB2, WSC2-LB, WSC2-LS, WSC2-LB2, MDS200-LB, MDS200-LS, MDS200-LB2, LDS40-LB, LDS40-LS, LDS40-LB2, LDS12-LB, LDS12-LS, LDS12-LB2, DDS04-LB, DDS04-LS, DDS04-LB2, RS485-LB, RS485-LB2, RS485-LS, RS485W-LB, RS485W-LS, RS485W-LB2) on reports and certificate. All the models are identical in electrically, the differences are shown in the table below. No other changes were made to them.

Model	Probes	PCB	Battery
DDS75-LB	Ultrasonic Probe	without charging module	8500mAh battery
RS485-LB	N/A		
SE01-LB	Soil Moisture Probe		
DDS45-LB	Ultrasonic Probe		
DDS20-LB	Ultrasonic Probe		
MDS120-LB	Microwave Probe		
MDS200-LB	Microwave Probe		
LDS12-LB	LiDar Probe		
SPH01-LB	Soil PH probe		
LMS01-LB	Leaf Moisture Probe		
DDS04-LB	Ultrasonic Probe		
LDS25-LB	LiDAR Distance Probe		
CS01-LB	Current Probe		
WQS-LB	Smoke Probe		
SE0X-LB	DR-SE-6P Soil Moisture & EC. Probe		
WSC2-LB	Alarm		
LDS40-LB	LiDAR Distance Probe		

DDS75-LS	Soil Moisture Probe	Has a charging module	3000mAh rechargeable battery
RS485-LS	without Probe		
SE01-LS	Ultrasonic Probe		
DDS45-LS	Microwave Probe		
DDS20-LS	Microwave Probe		
MDS120-LS	LiDar Probe		
MDS200-LS	Soil PH probe		
LDS12-LS	Leaf Moisture Probe		
SPH01-LS	Ultrasonic Probe		
LMS01-LS	Soil Moisture Probe		
DDS04-LS	Ultrasonic Probe		
LDS25-LS	LiDAR Distance Probe		
CS01-LS	Current Probe		
WQS-LS	Smoke Probe		
SE0X-LS	DR-SE-6P Soil Moisture & EC. Probe		
WSC2-LS	Alarm		
LDS40-LS	LiDAR Distance Probe		
DDS75-LB2	Soil Moisture Probe	Without charging module	19000mAh battery
RS485-LB2	without Probe		
SE01-LB2	Ultrasonic Probe		
DDS45-LB2	Microwave Probe		
DDS20-LB2	Microwave Probe		
MDS120-LB2	LiDar Probe		
MDS200-LB2	Soil PH probe		
LDS12-LB2	Leaf Moisture Probe		
SPH01-LB2	Ultrasonic Probe		
LMS01-LB2	Soil Moisture Probe		
DDS04-LB2	Ultrasonic Probe		
LDS25-LB2	LiDAR Distance Probe		
CS01-LB2	Current Probe		
WQS-LB2	Smoke Probe		
SE0X-LB2	DR-SE-6P Soil Moisture & EC. Probe		
WSC2-LB2	Alarm		
LDS40-LB2	LiDAR Distance Probe		
RS485W-LS	Without probe	1. Has a charging module 2. Support	3000mAh rechargeable battery

		12Vdc output	
RS485W-LB	Without probe	3. Has a charging module 4. Support 12Vdc output	8500mAh battery
RS485W-LB2	Without probe	1. Has a charging module 2. Support 12Vdc output	19000mAh battery

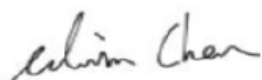
We confirm that all information above is true, and we'll be responsible for all the consequences. Please contact me if you have any question.

Sincerely Yours,

Signature

edwin chen

Manager



***** END OF REPORT *****